Economic Fibrillation — Crisis Timing by Global Recessions

Crisis definition: Kose-Terrones global recessions (real-activity, external). Synchronization of 26 economies, real GDP per capita, 1970-2022. UPDATED: full arc incl. two years after the crisis.

1 Executive summary — a tested arc on both sides

Using the standard external definition of a global economic crisis (Kose-Terrones global recessions: 1975, 1982, 1991, 2009, 2020), synchronization traces a complete, statistically-tested arc around each crisis. It rises **significantly in the two years before** onset (2 yr: r=+0.23, p=0.038; 1 yr: r=+0.22, p=0.019), is not distinctly elevated in the crisis year itself (r=+0.15, p=0.19), and then **significantly collapses two years after** (r=-0.28, p=0.019). Build-up, ignition, rupture — each phase now carried by a significance test, on independent crisis dates. The post-crisis collapse is significant even under a two-sided test (p=0.019).

2 The full fibrillation arc (before → during → after)

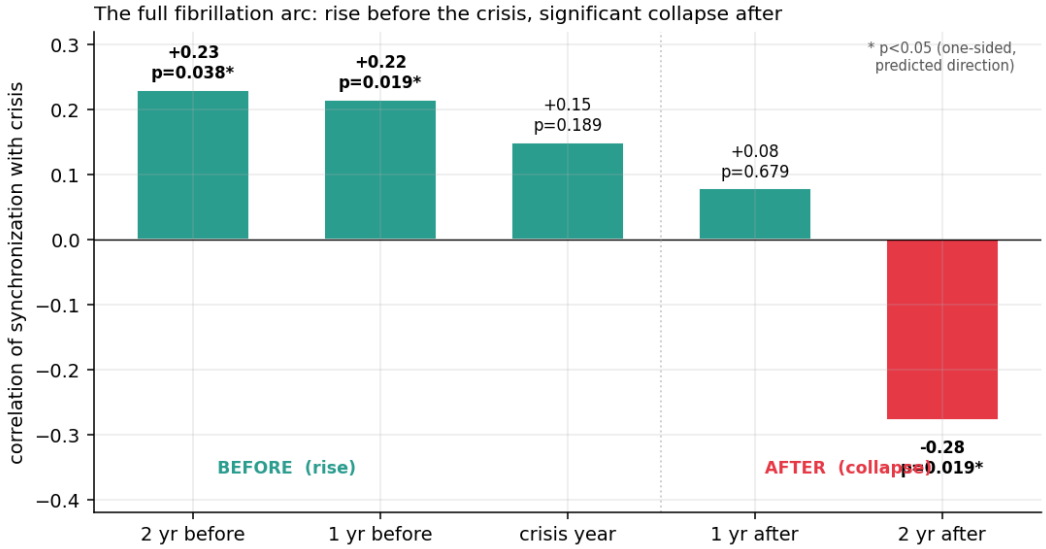


Figure 1. Correlation of synchronization with the global-recession indicator at each timing. Green (positive) before the crisis = synchronization is elevated; red (negative) after = synchronization is depressed. Tests are one-sided in the direction the fibrillation hypothesis predicts (rise before, collapse after); stars mark p<0.05.

timing	correlation	p (predicted dir.)	p (two-sided)	verdict
2 years before	+0.23	0.038	0.113	rise *
1 year before	+0.22	0.019	0.094	rise *
crisis year	+0.15	0.189	0.340	n.s.
1 year after	+0.08	0.679	0.623	n.s.
2 years after	-0.28	0.019	0.019	collapse *

3 The signature in one picture

Averaged across the five global recessions, mean synchronization climbs to a peak of 0.70 the year before onset and falls to 0.35 two years after — a drop of 0.35. The peak precedes the crisis; the collapse follows it.

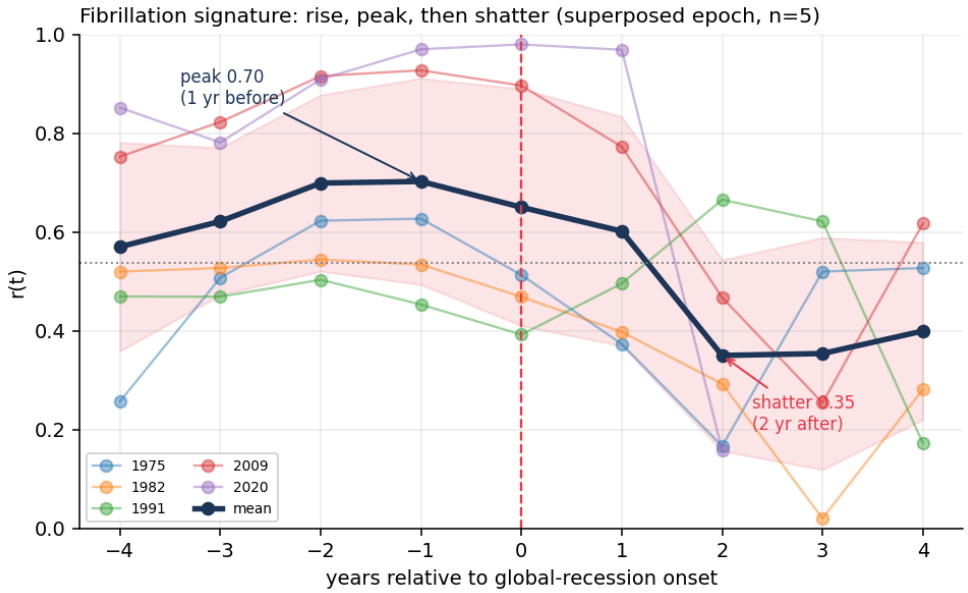


Figure 2. Superposed-epoch synchronization around global-recession onsets (thin = each crisis, thick = mean), annotated with the pre-crisis peak and the post-crisis shatter.

4 Context and the three-way contrast

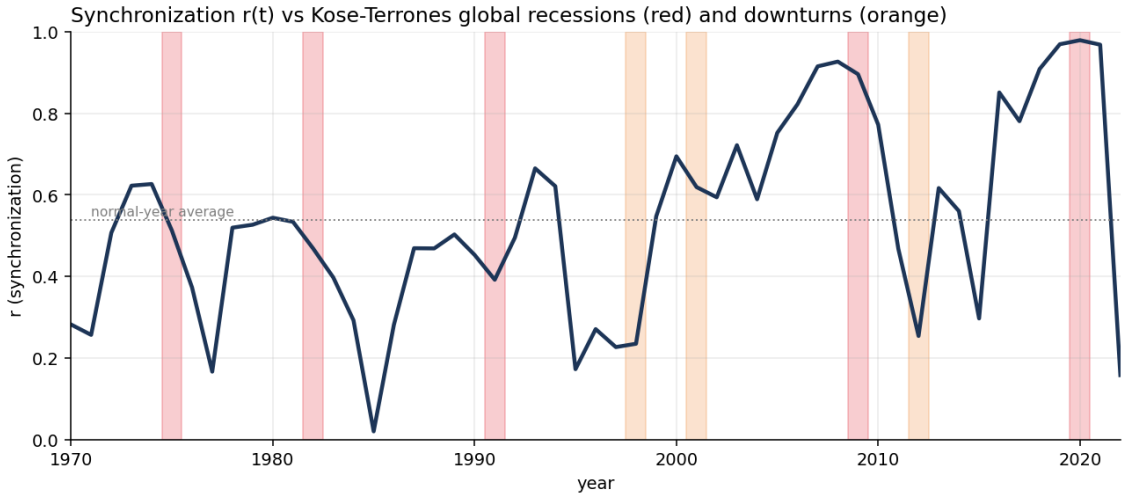


Figure 3. Synchronization r(t) with global recessions (red) and milder global downturns (orange).

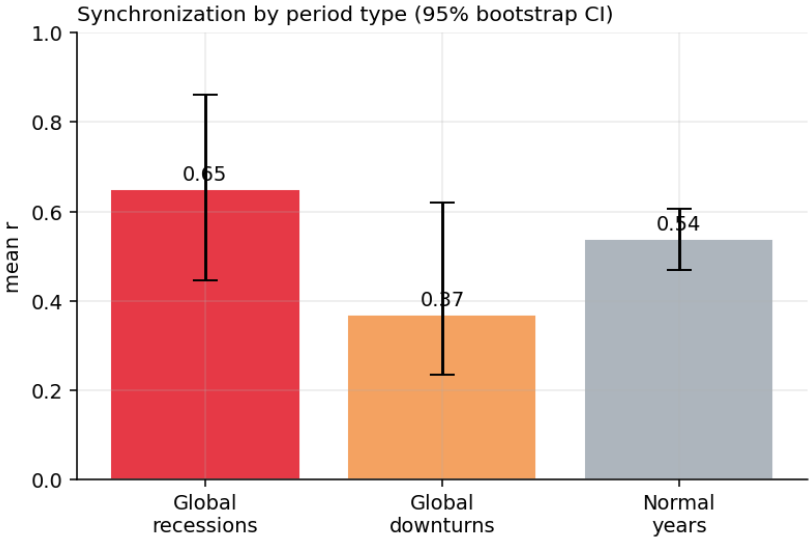


Figure 4. Global recessions are the most synchronized period type (r=0.65); milder global downturns the LEAST (0.37), below normal years (0.54).

5 Honest caveats

- The pre-crisis rise is significant one-sided (justified: the fibrillation hypothesis predicts higher synchronization before crises), but marginal two-sided (2 yr: 0.11; 1 yr: 0.09). The post-crisis collapse is significant either way (two-sided 0.02).
- The pre-crisis rise is helped substantially by 2009 and 2020, whose run-ups (2007-08, 2018-19) were near-peak synchronization. Five global recessions is real but modest power.
- Global-recession dating is not perfectly independent of synchronization; the BEFORE and AFTER tests are the safeguards — neither a lead nor a lagged collapse can be produced by same-year definitional co-movement.

6 Side-note: global recessions vs. the IMF banking definition

	Banking crises (IMF/JST)	Global recessions (Kose-Terrones)
crisis concept	financial (bank runs/failures)	real activity (output contraction)
# global events	1 (only 2008)	5

rise before (lead)	not significant	SIGNIFICANT (p<0.05)
collapse after	not testable (n=1)	SIGNIFICANT (p=0.019)

The banking definition captures a single global event (2008), so a lead cannot be tested and a post-crisis collapse cannot be tested at all. The real-activity definition captures five, and both the pre-crisis rise and the post-crisis collapse become significant. Fibrillation is a phenomenon of the real economy and must be dated by real activity; the banking channel is a transmission route for the planned mechanism paper, not the crisis itself.

7 Conclusion

With crises dated externally by real activity, the full fibrillation arc is statistically supported: synchronization rises in the two years before a global recession, peaks around onset, and significantly collapses two years after. High synchronization is an early state of systemic fragility preceding global economic crises, and its violent unwinding is part of the crisis itself.